04 — Important Queries

Item	Details
Course	DBMS Project — Semester 5
Team	Sneha Angelin Jisho (PES2UG24CS507); Sowmya Ramesh (PES2UG24CS512)
Stack	React + Vite; Node.js + Express; MySQL; MongoDB
Repository	https://github.com/snehajisho/Satellite_Data_Provenance_Earth_Observatory-DBMS_Project

10. Important Queries

This document summarises representative examples from `database/03_queries.sql` and `database/mongodb/queries.js`. The repository scripts remain the authoritative executable query listings.

10.1 MySQL query examples

Query	Purpose	SQL concept
Q1 Observation details	Join OBSERVATION, SENSOR, SATELLITE and GEO_LOCATION to display observation context.	Multiple INNER JOINs.
Q2 Satellites and sensors	List all satellites, retaining satellites with no sensor row.	LEFT JOIN.
Q3 Mission–satellite mapping	Show mission, satellite, role and assignment dates.	Three-table join through associative table.
Q4 Data product details	Show product, source observation and geographic area.	Relational joins.
Q5 Processing provenance	Resolve input and output product names for each activity.	Join DATA_PRODUCT twice.
Q6 Observation count by status	Count observations grouped by status.	GROUP BY and COUNT.
Q7 Average cloud cover by area	Calculate average cloud cover for each geographic area.	GROUP BY and AVG.
Q8 Product count by processing level	Count products for each processing level.	GROUP BY and COUNT.
Q9 Areas with multiple observations	Return areas having more than one observation.	GROUP BY and HAVING.

Query	Purpose	SQL concept
Q10 Repeated processing types	Return processing types with at least two activities.	GROUP BY and HAVING.
Q11 Above-average cloud cover	Find observations whose cloud cover exceeds the overall average.	Scalar subquery.
Further examples	The SQL source contains additional query examples beyond those listed above.	See `database/03_queries.sql` for full query text.

10.2 MongoDB query examples

Query	Purpose	MongoDB technique
Q1	Display all observation metadata documents.	find().pretty()
Q2	Retrieve a document by observation_id.	findOne().
Q3	Filter by satellite name such as Sentinel-2A.	Nested-field equality.
Q4	Filter by location such as Berlin.	Nested-field equality.
Q5	Find observations with cloud cover below 20%.	$lt comparison.
Q6	Find records with quality_flag GOOD.	Nested-field equality.
Q7	Find records containing a spectral band such as B08.	Array membership.
Q8	Find records where cloud masking is Available.	Nested-field filter.
Q9	Count observations per satellite.	Aggregation $group$/sum and $sort.
Q10	Calculate average cloud cover by location.	Aggregation $group$/avg.
Further examples	Additional aggregation examples are included in the MongoDB query script.	See `database/mongodb/queries.js`.

10.3 Stored routine examples

The implementation script demonstrates:

- `CALL GetObservationsBySatellite(1)`
- `CALL GetProductsByLocation(1)`
- `SELECT CountObservationsBySatellite(1) AS total_observations`